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Hamburg, [PLATZHALTER — Datum der Absendung]

[PLATZHALTER — Firma]
[PLATZHALTER — Ansprechpartner]
[PLATZHALTER — Straße]
[PLATZHALTER — PLZ, Ort]

Application as [PLATZHALTER — Stellenbezeichnung] — Reference [PLATZHALTER — Kennziffer]

Dear [PLATZHALTER — Ansprechpartner],

It is with genuine excitement that I am writing to apply for the position of [PLATZHALTER — Stellenbezeichnung] at [PLATZHALTER — Firma]. I have spent ten years building computational models and asking whether their outputs can be trusted — in physics, in ocean ecosystems, and most recently in clinical data. For the past months I have applied that same discipline to neural networks, and it is the work I want to do next.

After a Ph.D. in astrochemistry at Université Paris-Saclay, where I analysed terabytes of particle-accelerator data, I worked as a post-doctoral scientist at Universität Hamburg on marine ecosystem modelling. There I translated a legacy Fortran engine into a high-performance Python one using Google JAX with TPU/GPU parallelisation — my first serious encounter with the tensor operations that underpin modern deep learning frameworks.

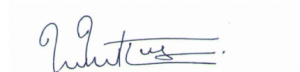
Since then I have been conducting independent research in neural network interpretability. I built a toolchain that opens up a 15-layer transformer (Leela Chess Zero) and reads out what it computes internally: the model is converted from ONNX to PyTorch, and forward hooks on every encoder layer capture its attention during inference. What I value most is not the tooling but what it taught me: I found two systematic errors in my own pipeline that produced entirely plausible results — neither raised an exception, neither failed a test, and one had already reached a published write-up before I caught it. I corrected it publicly: thejusmahajan.github.io/blog-lc0-attention-frame.html.

That habit is what I would bring beyond the tools themselves: a model returning confident, well-shaped, wrong answers is the failure mode that costs most and announces itself least. My work at HealthTwist on a national medical registry pipeline (143,000 patient records, some 300 clinics) rested on the same discipline — every refactoring step verified byte-identical before acceptance.

[PLATZHALTER — ein bis zwei Sätze zur konkreten Stelle: warum diese Firma, welches Projekt, welche Technologie aus der Anzeige.]

I hold work authorisation for Germany and am available immediately. My German is B1 (Goethe-Zertifikat), B2 in preparation, and I work daily in English. In September I begin the certified course “Deployable Data Analysis & AI Pipelines with HPC” at the Supercomputing-Akademie of the HLRS, Universität Stuttgart, building on my earlier training at the Jülich Supercomputing Centre. I would be glad to discuss how my background could serve your team, and happy to go into the interpretability work in whatever detail is useful.

Sincerely,



Dr. Thejus Mahajan